AN IMPURITY CAN BE REPLACED BY ONE WITH A LOCALLY FINITELY PRESENTED MORPHISM AND A CLOSED POINT OF THE FIBRE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex, lines 1-85) is a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project's own reference/slogan/history display-environment definitions copied mechanically from the shared preamble; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler, and extra_packages is left empty because everything needed is already in the retained preamble. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The situation and the notion of an impurity of a module above a point. Same chapter flat.tex, situation labelled situation-pre-pure (statement, proof line 4515). Fixes the data of an impurity that the proof is allowed to replace.
- Separation of a closed subset from a point, and base change of the relative assassin. Limits, label limits-lemma-separate (proof line 4525); Divisors, label divisors-lemma-base-change-relative-assassin (proof line 4552). Produces the open neighbourhood of the point, the locally finitely presented replacement morphism and the closed subscheme disjoint from the special fibre and containing the image of the closure, and transports the relative associated point to that replacement.
- Emptiness of the fibre of a closed set at a point spreads to an open subset of the closure, and fibres over a point are Jacobson schemes. More on Morphisms, label more-morphisms-lemma-empty-generic-fibre (proof line 4562); Morphisms, label morphisms-lemma-ubiquity-Jacobson-schemes (proof line 4566). Shows that, having assumed the morphism locally of finitely presented, the vanishing of the fibre of the closure at t persists on an open subset of the closure of t, and since the fibre over s is Jacobson it contains a closed point where the fibre still vanishes.

Source reference. Original lemma 05DC: sources/stacks-chow/tags.txt, source lines 6950–6950. Navigation pointer for \ref{divisors-lemma-base-change-relative-assassin} used inside the counted span (line 4552); the argument that consumes it is retained in the span.

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Source reference. Original lemma 05BD: sources/stacks-chow/tags.txt, source lines 6881–6881. Navigation pointer for \ref{limits-lemma-separate} used inside the counted span (line 4525); the argument that consumes it is retained in the span.

Source reference. Original lemma 054W: sources/stacks-chow/tags.txt, source lines 6654–6654. Navigation pointer for \ref{more-morphisms-lemma-empty-generic-fibre} used inside the counted span (line 4562); the argument that consumes it is retained in the span.

Source reference. Original lemma 02J6: sources/stacks-chow/tags.txt, source lines 3325–3325. Navigation pointer for \ref{morphisms-lemma-ubiquity-Jacobson-schemes} used inside the counted span (line 4566); the argument that consumes it is retained in the span.

Source reference. Original lemma 05FW: sources/stacks-flat/pinned-original-source.tex, source lines 4473–4473. Navigation pointer for \ref{situation-pre-pure} used inside the counted span (line 4515); the argument that consumes it is retained in the span.

Lemma 0.1. *In Situation 05FW. If there exists an impurity of $\mathcal{F}$ above s, then there exists an impurity $(g : T \to S, t' \rightsquigarrow t, \xi)$ of $\mathcal{F}$ above s such that g is locally of finite presentation and t a closed point of the fibre of g above s.*

Proof. Let $(g : T \to S, t' \rightsquigarrow t, \xi)$ be any impurity of $\mathcal{F}$ above s. We apply Limits, Lemma 05BD to $t \in T$ and $Z = \overline{\{\xi\}}$ to obtain an open neighbourhood $V \subset T$ of t, a commutative diagram

$$\begin{array}{ccc} V & \xrightarrow{a} & T' \\ \downarrow & & \downarrow b \\ T & \xrightarrow{g} & S, \end{array}$$

and a closed subscheme $Z' \subset X_{T'}$ such that

(1) the morphism $b : T' \to S$ is locally of finite presentation,
(2) we have $Z' \cap X_{a(t)} = \emptyset$, and
(3) $Z \cap X_V$ maps into Z' via the morphism $X_V \to X_{T'}$.

As t' specializes to t we may replace T by the open neighbourhood V of t. Thus we have a commutative diagram

$$\begin{array}{ccccc} X_T & \longrightarrow & X_{T'} & \longrightarrow & X \\ \downarrow & & \downarrow & & \downarrow \\ T & \xrightarrow{a} & T' & \xrightarrow{b} & S \end{array}$$

where $b \circ a = g$. Let $\xi' \in X_{T'}$ denote the image of ξ. By Divisors, Lemma 05DC we see that $\xi' \in \mathrm{Ass}_{X_{T'}/T'}(\mathcal{F}_{T'})$. Moreover, by construction the closure of $\overline{\{\xi'\}}$ is contained in the closed subset Z' which avoids the fibre $X_{a(t)}$. In this way we see that $(T' \to S, a(t') \rightsquigarrow a(t), \xi')$ is an impurity of $\mathcal{F}$ above s.

Thus we may assume that $g : T \to S$ is locally of finite presentation. Let $Z = \overline{\{\xi\}}$. By assumption $Z_t = \emptyset$. By More on Morphisms, Lemma 054W this means that $Z_{t''} = \emptyset$ for t'' in an open subset of $\overline{\{t\}}$. Since the fibre of $T \to S$ over s is a Jacobson scheme, see Morphisms, Lemma 02J6 we find that there exist a closed point $t'' \in \overline{\{t\}}$ such that $Z_{t''} = \emptyset$. Then $(g : T \to S, t' \rightsquigarrow t'', \xi)$ is the desired impurity. □